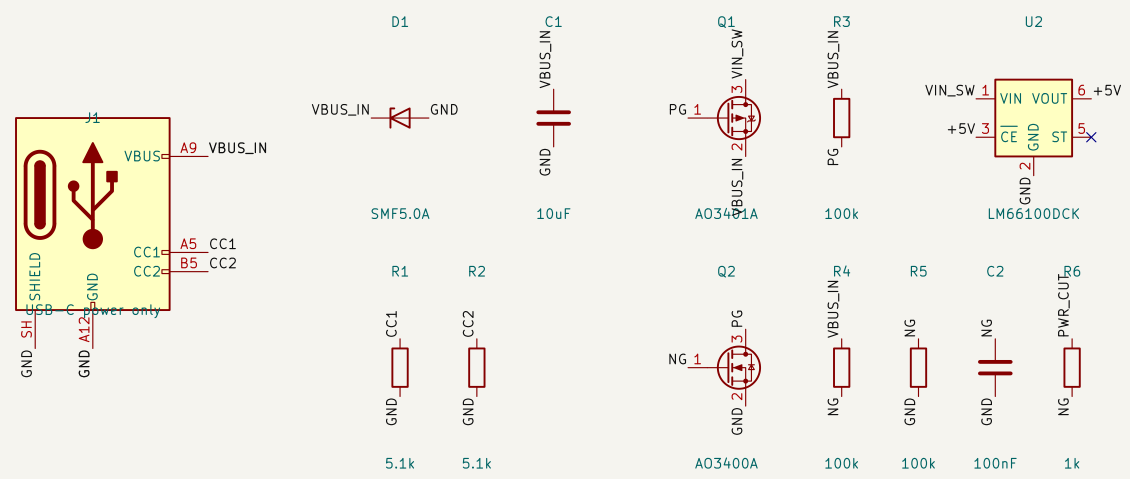
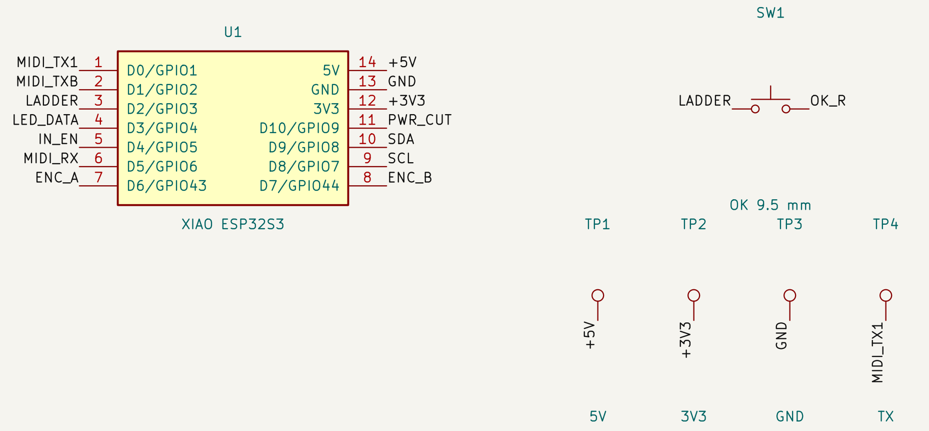
POWER IN: J1 power only (D+/D− not fitted). Q1 cut (on by default), U2 ideal diode onto the XIAO's 5V

U2 ~CE tied to VOUT: the datasheet's ideal−diode mode (blocks current back into J1). The XIAO's 5V pin IS its USB−C VBUS (no part between, Seeed v1.2 schematic): a computer on the XIAO port and a charger on J1 share +5V. Firmware pulls PWR_CUT low once a computer has enumerated it (device mode): Q2 off, Q1 off, J1 cut. At reset PWR_CUT floats: R4/R5 hold Q2 on, so a charger alone always starts it.
C4 (100 uF) not fitted: host−mode hot−plug reserve, against USB's 10 uF limit for a device.

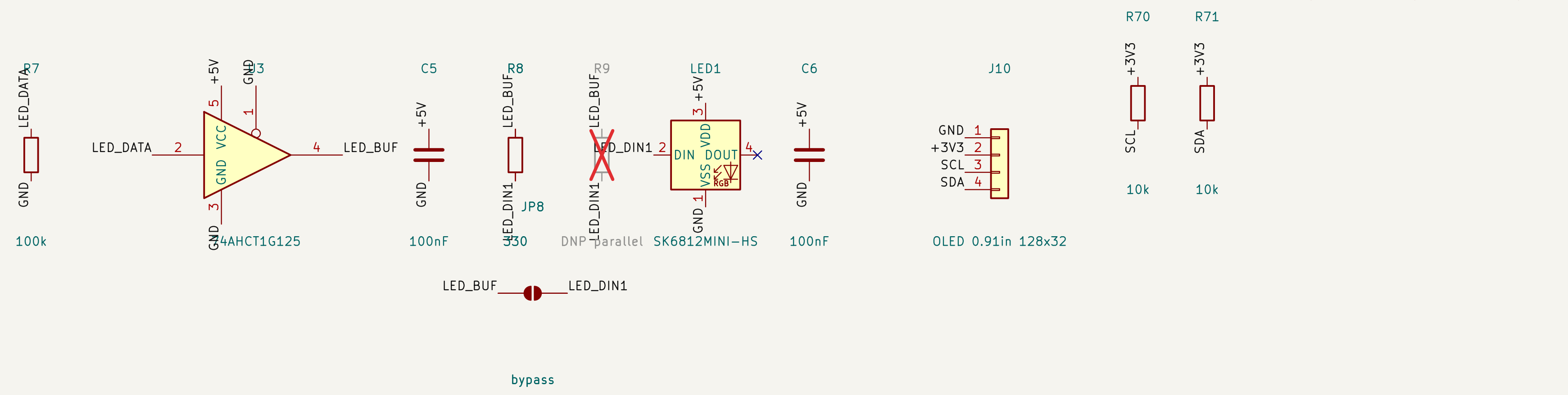


XIAO ESP32S3 (Seeed footprint, 2 x 7 headers or castellations)

D6 = GPIO43 has 499 ohm on the module (U0TXD). D2 = GPIO3 strapping, ignored by default eFuses. ENC_A = D6 = GPIO43, ENC_B = D7 = GPIO44 (U0TXD / U0RXD), IN_EN = D4 = GPIO5, MIDI_RX = D5 = GPIO6 (v5).
v5: MIDI_TX1 = D0 = GPIO1 (OUT1 alone: Hi−Z in IN mode); MIDI_TXB = D1 = GPIO2 drives OUT2, OUT3 and OUT4 together (3 loops on one pad).
LADDER = D2 = GPIO3 (ADC1): OK / BACK / PUSH on one resistor ladder. LED_DATA = D3 = GPIO4. SCL = D8 = GPIO7, SDA = D9 = GPIO8 (the OLED, I2C by the GPIO matrix). PWR_CUT = D10 = GPIO9, low = J1 cut.



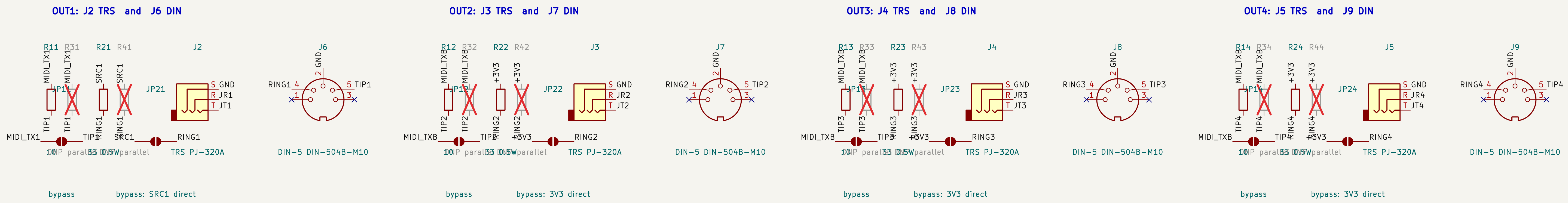
LED: one SK6812MINI−HS on +5V (top); U3 lifts the 3.3 V data to 5 V (VIH = 0.7 x VDD = 3.5 V at 5 V). SCREEN: J10, a 0.91 in 128 x 32 OLED (SSD1306, I2C 0x3C, 3V3)

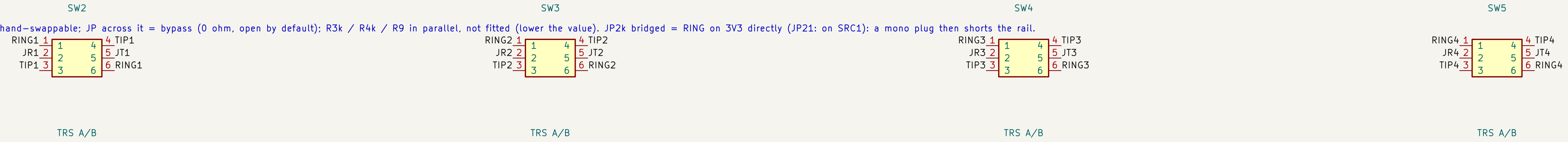


On the board each pair is ONE footprint (J6..J9, jank:MIDI_OUT_PJ−320A_top_DIN−504B−M10_bottom: the jack's pads T, R, S carry J2..J5's nets, its switch pin SW open; the DIN's pins on the bottom). J2..J5 are BOM lines only.

MIDI OUT 1−4 (CA−033 at 3.3 V), v5: BOTH fitted on every output, the TRS (type A, top) AND the DIN−5 (bottom), in parallel; plug one at a time

TRS type A: tip = data (10 ohm to the TX pad), ring = source (33 ohm 0.5 W to 3V3), sleeve = GND. DIN−5: pin 5 = tip net, pin 4 = ring net, pin 2 = GND, pins 1 and 3 open.





Each MIDI resistor (and R8): axial, hand−swappable; JP across it = bypass (0 ohm, open by default): R3k / R4k / R9 in parallel, not fitted (lower the value). JP2k bridged = RING on 3V3 directly (JP21: on SRC1): a mono plug then shorts the rail.

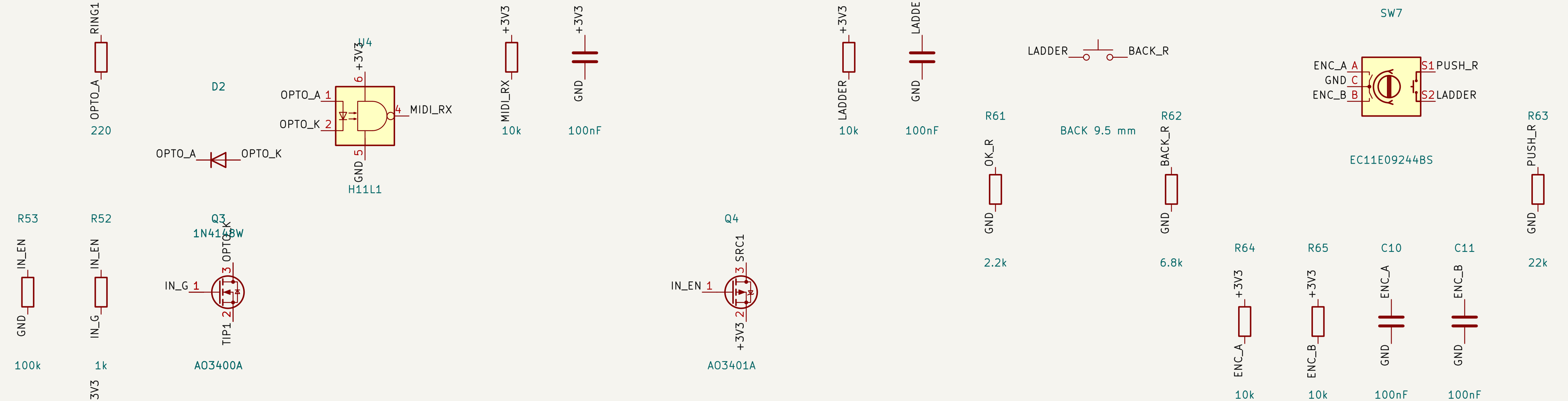
SW2..SW5: TRS type A / B per output (top side): they swap the jack's T and R between TIPk (data, sink) and RINGk (source) after the resistors; the DIN keeps pin 5 = TIPk, pin 4 = RINGk. 1−2 / 4−5 closed = type A.

OUT1 AS MIDI IN (software): TX1 to input (Hi−Z), IN_EN high (Q4 off: no source), Q3 on), MIDI_RX by UART RX

CONTROLS (top side, by the screen): one ADC ladder on GPIO3: R60 10k to 3V3; OK 2.2k, BACK 6.8k, PUSH 22k to GND

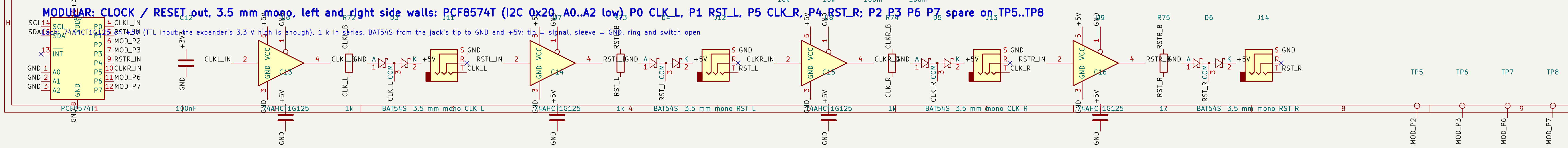
OUT mode: IN_EN low (R53 at reset): Q4 on (3V3 to the 33 ohm), Q3 off, the opto's loop open (its body diode blocks). [illegible]

Idle 3.30 V, OK 0.60 V, BACK 1.34 V, PUSH 2.27 V [illegible] Encoder A/B: 10k to 3V3, 100 nF to GND each.



MODULAR: CLOCK / RESET out, 3.5 mm mono, left and right side walls: PCF8574T (I2C 0x20, A0..A2 low) P0 CLK_L, P1 RST_L, P5 CLK_R, P4 RST_R; P2 P3 P6 P7 spare on TP5..TP8

Each: 74AHCT1G125 on +5V (TTL input: the expander's 3.3 V high is enough), 1 k in series, BAT54S from the jack's tip to GND and +5V; tip = signal, sleeve = GND, ring and switch open



generated by make_schematic.py; do not edit by hand
hardware/pcb/README.md: the power path, the pin table and every unverified item

Sheet: /
File: jank−pcb−mod.kicad_sch

Title: jank: XIAO ESP32S3 MIDI clock stick

Size: A2
Date: 2026−11−08
Rev: 0.1 schematic for review
KiCad E.D.A. 10.0.8
Id: 1/1